27.09.2022

Book: Digital Systems by Ronald J. Tocci

Binary signal → 2 টি discrete value

0 ~ 0.8 V = 0 logic
2.4 ~ 5 V = 1 logic
0.8 ~ 2.4 V = forbidden region

} Typical value

forbidden region ↓, performance ↓

chip → square / rectangular to lessen the costing

Weighted number → customized base

10.10.2022

logic operation → Basic gate

OR +

AND ·

Decimal fraction এ last bit 5 or 0 হলে, finite binary equivalent পাওয়া যাবে।

★ $(3.125)_{10} = (?)_2$

★ Algorithm & flow chart of conversion.

Complement's

1's complement
2's "
⋮
r'th "

1's complement

$101011 \xrightarrow{1's} 010100$

bit synchronization এর মাধ্যমে প্রতিটি NOT Gate দিয়ে হয়:

Truth Table → Boolean expression → Circuit

<u>2's complement</u>

lsb of 1's complement + 1

$$101011 \longrightarrow \begin{array}{r} 010100 \\ +1 \\ \hline 010101 \end{array}$$

$10101 \xrightarrow{2's} 01011$ [lsb কে ঠিক রেখে বাকি বিটে 1's complement]

$$\begin{array}{r} 10101 \\ \downarrow 1's \\ 01010 \\ +1 \\ \hline 01011 \end{array}$$

So, this process is not universal.

$10100 \xrightarrow{2's} \underline{01010}$
Not correct

$$\begin{array}{r} 10100 \\ \downarrow 1's \\ 01011 \\ +1 \\ \hline 01100 \end{array}$$

<u>General process</u>: lsb থেকে msb এর দিকে যেতে যেতে যতক্ষণ না 1 পাওয়া যাবে, ততক্ষণ 1's complement apply করা যাবে না।

$10\boxed{100} \rightarrow$ Intact

$\downarrow 2's$

01100

<u>H.W.</u> r th complement

2's complement দিয়ে negative number পাওয়া যায়। $9 \xrightarrow{2's} -9$

sin	mag
+	10

MSB → sign

1 → negative

0 → positive

Total combinations: 2^n [n = no. of bit]

2's complement form for signed numbers:

$$-(2^{n-1}) \text{ to } +(2^{n-1}-1)$$

$n = 8$, $-(2^{8-1}) = -128$ (minimum)

$+(2^{8-1}-1) = +127$ (maximum)

$1k = 2^{10}$

$10000000 = 128$ (unsigned)

$= -128$ (signed)

MSB = 1 & other bits = 0 হলে, signed & unsigned এর binary representation same হয়।

signed $\xrightarrow{2's}$ magnitude

Universal ছাড়া শুধু BCD তে 4 bit use করা হয়।

★BCD to Binary

<u>BCD Addition</u>

```
369 → 0011 0110 1001
245 → 0010 0100 0101
---   --------------
614   0101 1010 1110
 ↓     5    10   14   → representation
                        9 এর চেয়ে বড় হলে,
0110 0001 0100   0110 0110   correction needed
              --------------
              0110 0001 0100
               6    1    4
```

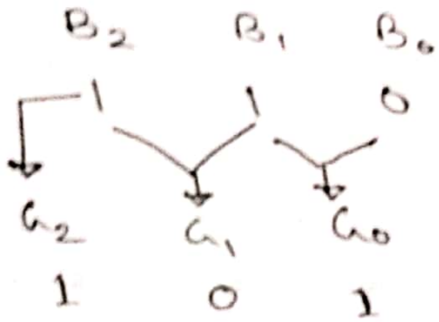
highest possible using 4 bit = 1111
but highest possible in BCD, 9 = 1001

sub: 0110 → 6

That is why, Addition এর পর 9 এর বেশি হয়ে গেলে, যে যে bit হয় হবে, তার সাথে 0110 add করতে হবে।

17.10.2022

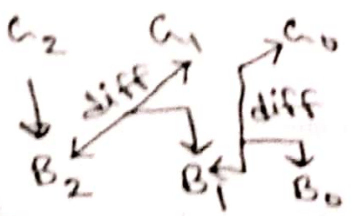
Binary to Gray → দুটি consecutive bit এর difference & MSB কেটর রেখে- রাখবে।



B_1	B_0	Diff
0	0	0
1	0	1
0	1	1
1	1	0

which turns out to be an Ex-OR Gate principle.

Gray to Binary → প্রতিটি output কে input হিসেবে use করতে হবে।



ASCII Code → basic: 7 bit
used to: 8 bit

application
(keyboard এ)

Error detection এর জন্য parity bit ব্যবহার। অর্থাৎ code mainly transmission এর জন্য use করা হয়।

Parity bit
- → odd (0, 1)
- → Even (0, 1)

7	6	5	4	3	2	1	0
1	0	0	1	1	0	0	1

even parity

Even method: যতগুলো 1 আছে, তাই even করতে জন্য 1 বসবে।

যদি even number of 1 থাকে, তবে parity bit 0 হবে।

18.10.2022

Logic 0 (Typical 0 ~ 0.8 V)

logic 1 (Typical 2.4 ~ 5 V)

variable → 2 → 2^2 entries

variable → n → 2^n entries

$$\left.\begin{aligned} X &= AB \\ X &= A.B \end{aligned}\right\} \text{AND}$$

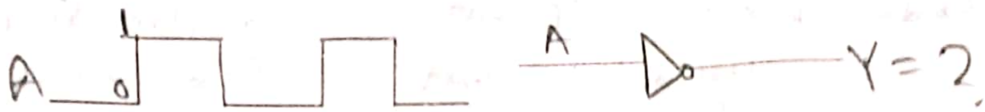
$$\#1 + xy + yz + p = 1 + Q = 1$$

$1 . X = X$

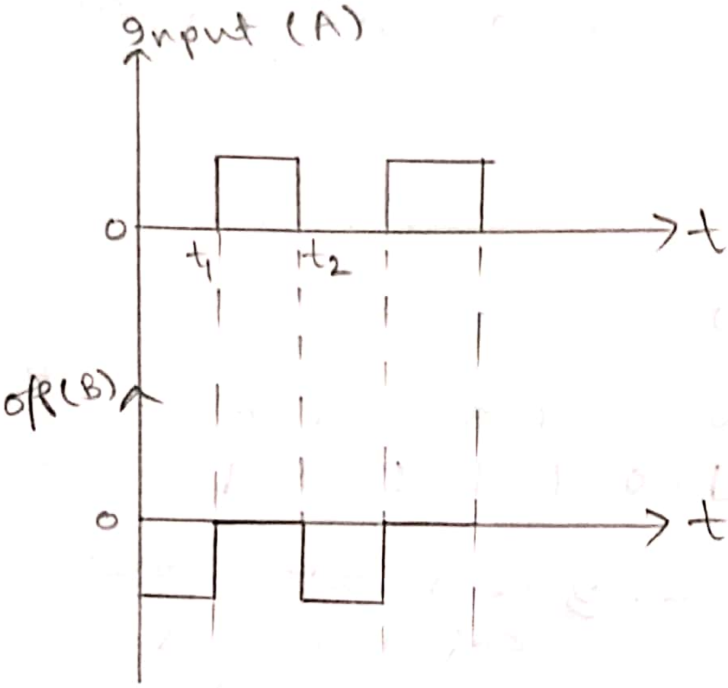
$$\boxed{\begin{aligned} x + \bar{x} &= 1 \\ x . \bar{x} &= 0 \end{aligned}}$$

CamScanner

#

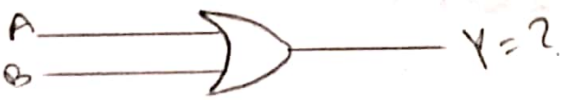


A
1
0
A
Y = ?

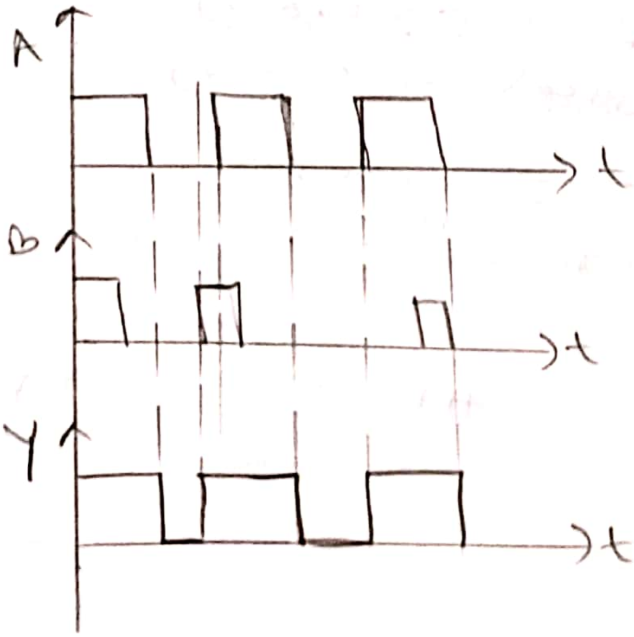


Input (A)
0
t
t_1
t_2
o/p (B)
0
t

#

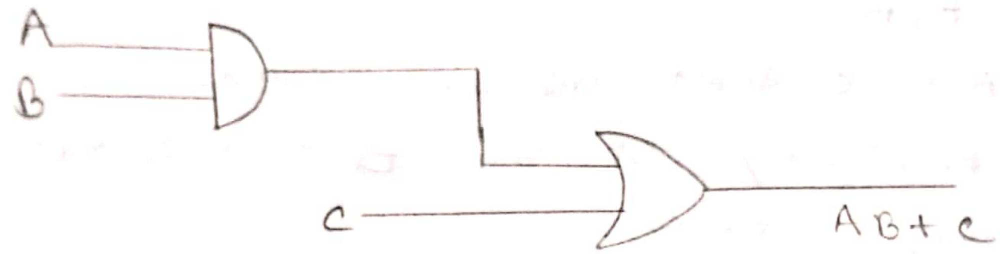


A
B
Y = ?



A
t
B
t
Y
t

#



When will be the circuit activated?

→

$A=1$

$B=1$

$AB=1$ $\qquad AB+C=1+0=1$

$C=0,1$ $\qquad AB+C=1+1=1$

i. $A=1,\ B=1,\ C=0$

$A=1,\ B=1,\ C=1$

ii. $A=0,\ B=1,\ AB=0,\ C=1,\ AB+C=0+1=1$

$A=1,\ B=0,\ AB=0,\ C=1,\ AB+C=1$

Input			Output
A	B	C	Y=AB+C
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

i. C → high

ii. A & B → both high

Fan A, B, C

i) A, B, C don't ON at a time

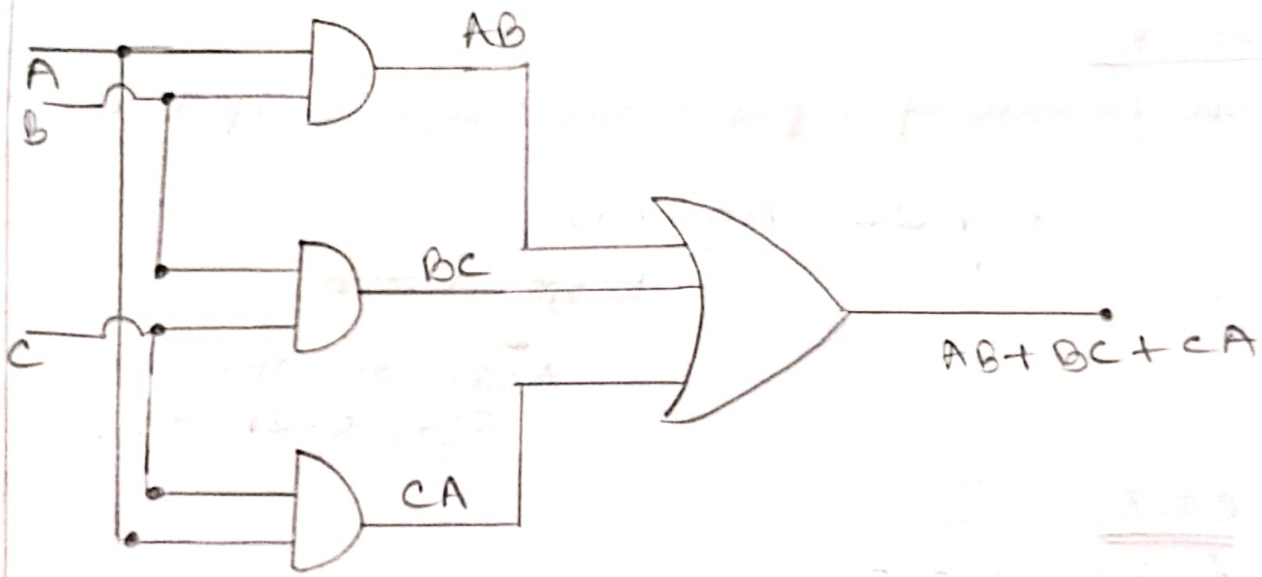
ii) A always ON but B and C not ON at a time.

Input			Output	
A	B	C	Y	
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	1

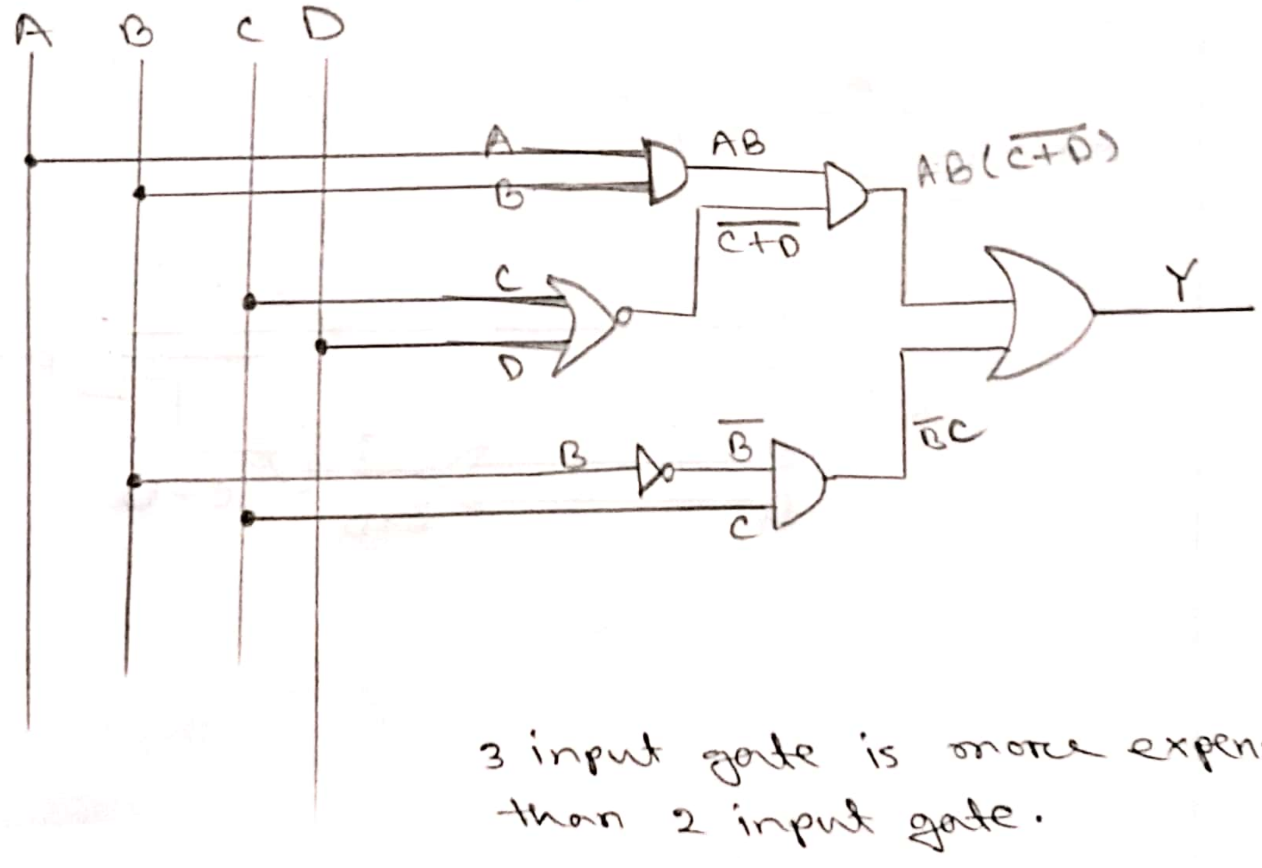
i) Three input circuits

ii) Circuit Activates when majority inputs are activated.

#



$Y = \underbrace{AB(\overline{C+D})}_{P_1} + \underbrace{\overline{B}C}_{P_2}$



3 input gate is more expensive than 2 input gate.

NAND
NOR } Universal Gate

Ex-OR

ODD number of input active থাকলে, output active.

$$Y = A \oplus B = \bar{A}B + A\bar{B}$$

→ minterm
↓
Activate term এর জন্য লেখা হয়।

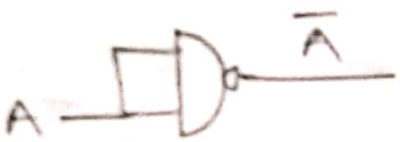
CT-01

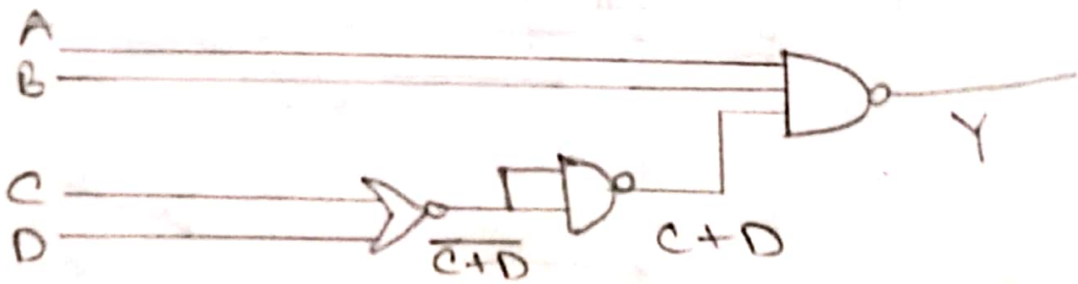
Lecture: 1, 2, 3

24.10.2022

$Y = \overline{AB.(C+D)}$

Implement by NOR and NAND gate.





$Y = \overline{AB\,\overline{(C+D)}}$, only NOR gate.

Boolean Theorem is not vector dependent.

$Y = a + \bar{a}b$. simplify the expression.

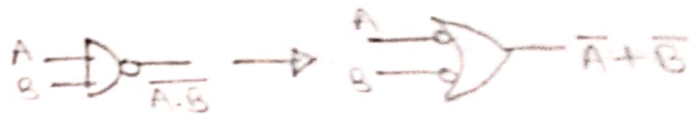
$$
\begin{aligned}
Y &= a + \bar{a}b \\
&= a.1 + \bar{a}b \\
&= a(1+b) + \bar{a}b \\
&= a + ab + \bar{a}b \\
&= a + b(a+\bar{a}) \\
&= a + b.1 \\
&= a + b
\end{aligned}
$$

$$
\boxed{\begin{aligned} 1+x &= 1 \\ x+\bar{x} &= 1 \\ x.\bar{x} &= 0 \\ x.x &= x \end{aligned}}
$$

$$\boxed{a + \bar{a}b = a + b}$$

De Morgan's Theorem

i. $\overline{x.y} = \bar{x} + \bar{y}$

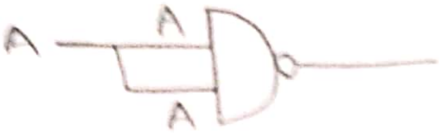


ii. $\overline{x+y} = \bar{x}.\bar{y}$

NAND, NOR — universal gate

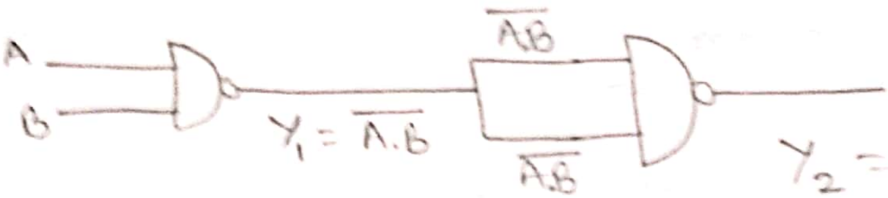
NAND Gate as Universal Gate

1. NAND Gate as NOT



$Y = \overline{A.A} = \overline{A}$

2. NAND Gate as AND

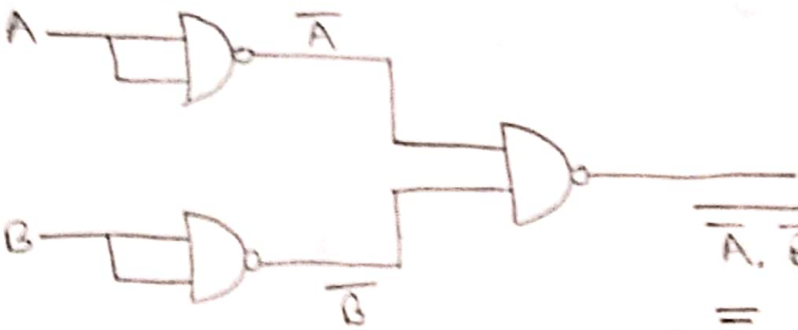


$Y_2 = \overline{\overline{AB}.\overline{AB}}$

$= \overline{\overline{AB}}$

$= AB$

3. NAND Gate as OR



$\overline{\overline{A}.\overline{B}}$

$= \overline{\overline{A}} + \overline{\overline{B}}$

$= A + B$

Inversion bubble

Active-high এর power dissipation বেশি।
That's why Active-low এর ব্যবহার বেশি preferable (i.e., NOR, NAND Gate দিয়ে)।

#

$$\overline{A}+\overline{B} = \overline{\overline{\overline{A}+\overline{B}}} = \overline{\overline{\overline{A}}\,.\,\overline{\overline{B}}} = \overline{A.B}$$

Canonical & standard forms

$Y = \overline{A}B + AB \rightarrow$ sum of product (SOP) → m

$Y = (A+B).(A+\overline{B}) \rightarrow$ Product of sum (POS) → M

canonial form → minimize করার পর যে form আসবে।

variable → 0 → product term এ → minterm → complementary

→ 1 → ~~product~~ sum → maxterm → complementary

x	y	z	Minterm	maxterm
0	1	1	$x'yz$	$x+y'+z'$

$Y_1 = x'y'z' + xyz$

$\quad\;\; 0\;0\;0 \quad\; 1\;1\;1$

$= m_0 + m_7$

$Y_1(x, y, z) = \Sigma\{0, 7\}$ → Function Representation

$Y_2 = (x + y + z)(x' + y' + z')$

$= M_0 . M_7$

$Y_2(x, y, z) = \Pi(0, 7)$

Variable কে minterm / maxterm এ প্রকাশ করার পর, decimal এ convert করতে হবে। আর এই decimal - টাই m/M এর suffix.

m = minterm

M = Maxterm

Σ = for m

Π = for M

#

A	B	C	F	Min(F)	Max(F)
0	0	0	1	$\bar{A}\bar{B}\bar{C}$	
0	0	1	1	$\bar{A}\bar{B}C$	
0	1	0	0		$A+\bar{B}+C$
0	1	1	1	$\bar{A}BC$	
1	0	0	1	$A\bar{B}\bar{C}$	
1	0	1	0		$\bar{A}+B+\bar{C}$
1	1	0	1	$AB\bar{C}$	
1	1	1	0		$\bar{A}+\bar{B}+\bar{C}$

$$F = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + AB\bar{C}$$

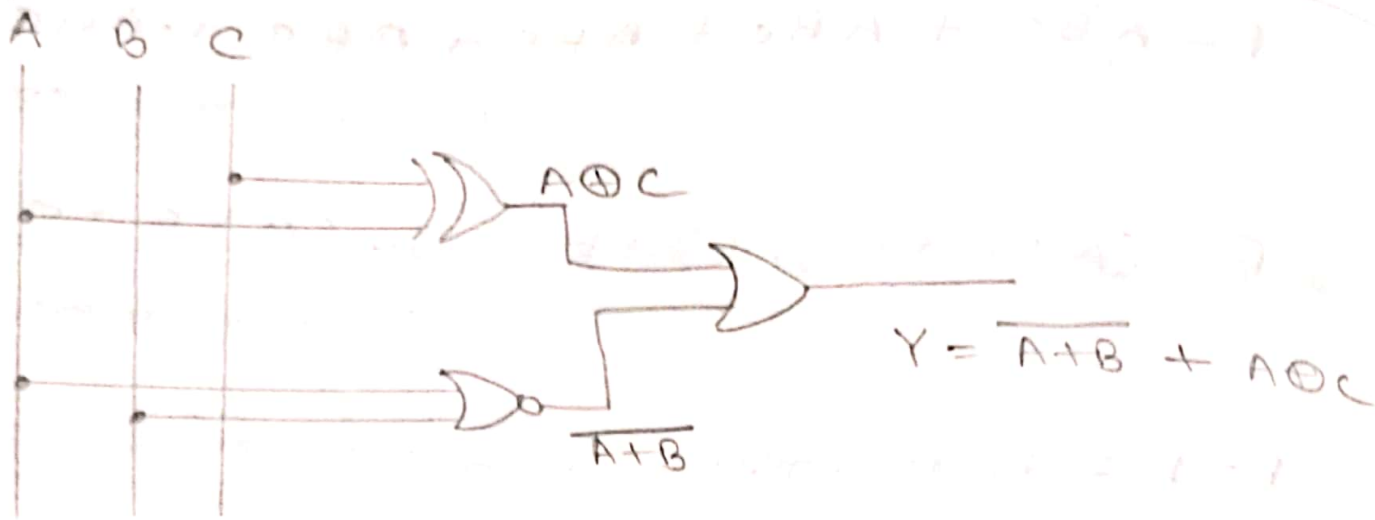
(using minterm)

$$F = (A+\bar{B}+C).(\bar{A}+B+\bar{C}).(\bar{A}+\bar{B}+\bar{C})$$

(using maxterm)

$F=1$ হলে, minterm এ প্রকাশ করতে হবে
$F=0$ " , maxterm " " "
But দুইটাই করতে হবে না একসাথে, যেকোনো একটির ওপরে করলেই হবে। এরপর simplify করতে হবে। maxterm এর simplification করা tough.

$$F_{min} = \bar{A}\bar{B}(\bar{C}+C) + A\bar{C}(\bar{B}+B) + \bar{A}BC$$
$$= \bar{A}\bar{B} + \bar{A}BC + A\bar{C}$$
$$= \bar{A}(\bar{B}+BC) + A\bar{C}$$
$$= \bar{A}(\bar{B}+C) + A\bar{C}$$
$$= \bar{A}\bar{B} + \bar{A}C + A\bar{C}$$
$$= \bar{A}\bar{B} + A \oplus C$$
$$= \overline{\overline{\bar{A}\bar{B}}} + A \oplus C$$
$$= \overline{\bar{\bar{A}} + \bar{\bar{B}}} + A \oplus C$$
$$= \overline{A+B} + A \oplus C$$



$Y = AB + BC + CA$

$= AB(C+\bar{C}) + BC(A+\bar{A}) + CA(B+\bar{B})$

$= ABC + AB\bar{C} + ABC + \bar{A}BC + ABC + A\bar{B}C$

$= ABC + AB\bar{C} + \bar{A}BC + A\bar{B}C$

$= m_7 + m_6 + m_3 + m_5$

$Y(A,B,C) = \Sigma(3,5,6,7)$

standard form contains all the variables.

$Y = A(B+C)$

$= (A+0+0).(0+B+C)$

$= (A+B\bar{B})(A\bar{A}+B+C)$

$= (A+B)(A+\bar{B})(A+B+C)(\bar{A}+B+C)$

$= (A+B+C\bar{C})(A+\bar{B}+C\bar{C})(\bar{A}+B+C)(A+B+C)$

$= (A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(A+\bar{B}+\bar{C})$
$(A+B+C)(\bar{A}+B+C)$

$$= (A+B+C)(\bar{A}+B+C)(A+\bar{B}+C)(A+B+\bar{C})(A+\bar{B}+\bar{C})$$

$$Y(A,B,C) = \Pi(0,1,2,3,4)$$

$Y = A + BC$

generate the function for maxterm.
(though it's in minterm হয়)

→ minterm বের করে, বাকি যতগুলো term গুলোই আসবে maxterm.

$$Y = A + BC = A(B+\bar{B}) + BC(A+\bar{A})$$
$$= AB + A\bar{B} + ABC + \bar{A}BC$$
$$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + ABC + \bar{A}BC$$
$$= \underbrace{ABC + \bar{A}BC + A\bar{B}C + AB\bar{C} + A\bar{B}\bar{C}}_{\text{minterm}}$$

$$\therefore Y(A,B,C) = \Sigma(3,4,5,6,7)$$

$$\therefore Y(A,B,C) = \Pi(0,1,2)$$

$$\Rightarrow Y = \underbrace{(A+B+C).(A+B+\bar{C}).(A+\bar{B}+C)}_{\text{maxterm}}$$

Karnaugh Map / K-Map

31.10.2022

Karnaugh Maps (K-maps)

$Y = ABC + \bar{A}BC + A\bar{B}C + AB\bar{C}$
$= AB + BC + CA$
$= EX\text{-}OR$

> Both are correct

k-map দিয়ে optimal solution পাওয়া যাবে।

2-variable k-maps

$AB + A\bar{B}$
$= A(B + \bar{B})$
$= A$

no. of var = 2
$\therefore$ sub cells $= 2^2 = 4$

যতগুলো combination হবে, ততগুলো subcell হবে k-map এ।

subcell $= 2^n$
n = no. of variable

A \ B	0	1
0	00	01
1	10	11

mapping of 2-var

A \ BC	00	01	11	10
0	000	001	011	010
1	100	101	111	110

Mapping of 3-var.

subcell, $2^3 = 8$

Input output

A	B	Y=A+B
0	0	0
0	1	1
1	0	1
1	1	1

variable = 2

sub-cell $= 2^2 = 4$

		B = 0	B = 1
A	0	0 (00)	1 (01)
	1	1 (10)	1 (11)

একটির loop এ যতগুলো 1 আসতে পারে $\rightarrow 2^n$

$= 2^{n(0,1,2)}$

$= 2^0, 2^1, 2^2$

$= 1, \underline{2}, 4$

4 ↳ invalid as we have only 3 1s here.

2 → সর্বোচ্চ 2টি আসবে এই কেসে

$\therefore f(A,B) = A+B$

#

		A = 0	A = 1
BC	00	0	1
	01	0	1
	11	0	1
	10	0	1

max No. of one in a loop,

$= 2^3$

$= 2^0, 2^1, 2^2, 2^3$

2^3 ↳ invalid

→ 1s should be adjacent.

→ loop continues until one 1 is outside the loop.

3-variable k-maps

#

BC \ A	0	1
00	1	1
01	0	1
11	0	0
10	1	1

open loop এ AB change হচ্ছে, but c fixed থাকছে।

$c = 0$

So, এটি 4 টি 1 adjacent.

$\therefore Y = \bar{C}$

#

BC \ A	0	1
00	1	1
01	0	1
11	0	0
10	1	1

→ এটি single হয়ে যায়

So, একে $2^1 = 2$ টি adjacent নিয়ে loop করা।

$\therefore A\bar{B}$

যদি 2 টি adjacent ও না থাকত, তবে $2^0 = 1$ টি one নিয়েই করতে হতো।

$\therefore Y = \bar{C} + A\bar{B}$

$F = \Sigma(0,1,3,7)$ → minimum 3 variable এর map হবে।

AB \ C	0	1
00	1 (000)	1 (001) → $\bar{A}\bar{B}$
01	0 (010)	1 (011) → BC
11	0 (110)	1 (111)
10	0 (100)	0 (101)

BC \ A	0	1
00	1 (000) → $\bar{A}\bar{B}$	0 (100)
01	1 (001)	0 (101)
11	1 (011)	1 (111) → BC
10	0 (010)	0 (110)

$Y = \bar{A}\bar{B} + BC$ $\qquad$ $Y = \bar{A}\bar{B} + BC$

2 নিয়মেই same answer আসছে। So, k-map is giving the optimal solution.

01.11.2022

#

BC \ A	0	1
00	1	0
01	1	1
11	1	0
10	0	0

k-map

minterm,

$F = \bar{A}\bar{B} + \bar{A}C + \bar{B}C$

$F(a,b,c) = \bar{a}c + abc + b\bar{c}$

↑ MSB unless otherwise stated.

$= \bar{a}c(b+\bar{b}) + abc + b\bar{c}(a+\bar{a})$

$= \bar{a}bc + \bar{a}\bar{b}c + abc + ab\bar{c} + \bar{a}b\bar{c}$

$= \Sigma(3,1,7,6,2)$

$= \Sigma(1,2,3,6,7)$

BC \ A	0	1
00	0 (0)	0 (4)
01	1 (1)	0 (5)
11	1 (3)	1 (7)
10	1 (2)	1 (6)

k-map

$F = B + \bar{A}C$

4 টার Adjacent 1 এর জন্য A আর C এর value variable হওয়ায়, only B.
আর 2 টা 1 এর জন্য শুধু B change হচ্ছে। তাই $\bar{A}C$.

#

		A = 0	A = 1
BC	00	1	0
	01	0	1
	11	1	0
	10	0	1

$$F = \bar{A}\bar{B}\bar{C} + A\bar{B}C + \bar{A}BC + AB\bar{C}$$

k-map এ complex gate এর minimization করতে পারে না।

$$F = (\bar{A}\bar{B}\bar{C} + \bar{A}BC) + (A\bar{B}C + AB\bar{C})$$

$$= \bar{A}(\bar{B}\bar{C} + BC) + A(\bar{B}C + B\bar{C})$$

$$= \bar{A}(\overline{B \oplus C}) + A(B \oplus C)$$

$$= \bar{A}\bar{x} + Ax \quad \Big| \; B \oplus C = x$$

$$= \overline{A \oplus x}$$

$$= \overline{A \oplus (B \oplus C)}$$

Design a full Adder.

Input			Output	
A	B	C	C_0	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

K-map for carry:

BC \ A	0	1
00	0_{0}	0_{4}
01	0_{1}	1_{5}
11	1_{3}	1_{7}
10	0_{2}	1_{6}

$$\text{carry} = BC + AC + AB$$

K-map for sum:

BC \ A	0	1
00	0_{0}	1_{4}
01	1_{1}	0_{5}
11	0_{3}	1_{7}
10	1_{2}	0_{6}

$$\text{Sum} = A\bar{B}\bar{C} + \bar{A}\bar{B}C + ABC + \bar{A}B\bar{C}$$

$$Sum = A(\bar{B}\bar{C} + BC) + \bar{A}(\bar{B}C + B\bar{C})$$

$$= A(\overline{B \oplus C}) + \bar{A}(B \oplus C)$$

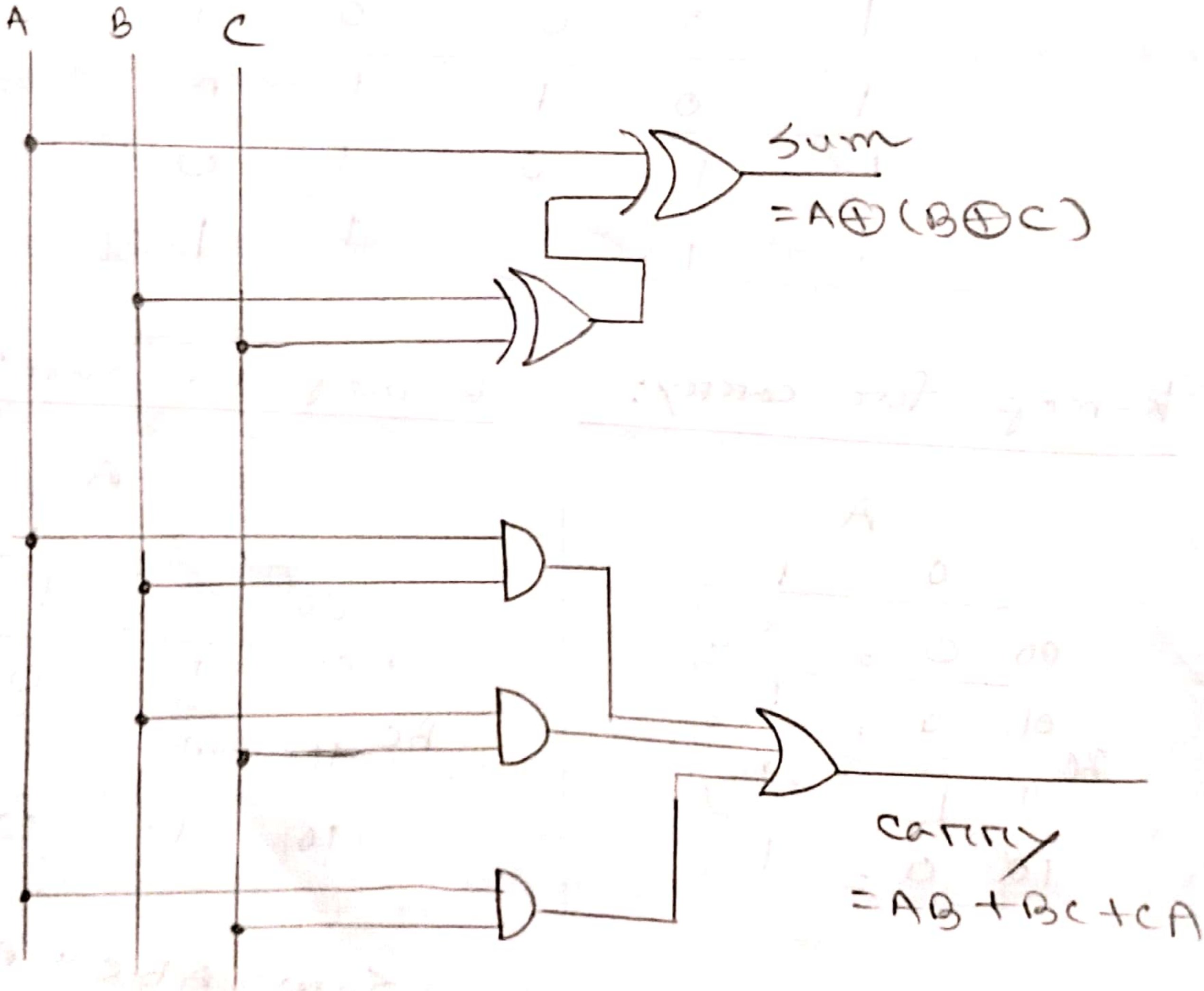
$$\cancel{= A \oplus (B \quad A \oplus}$$

$x = B \oplus C$

$$= A.\bar{x} + \bar{A}.x$$

$$= A \oplus x$$

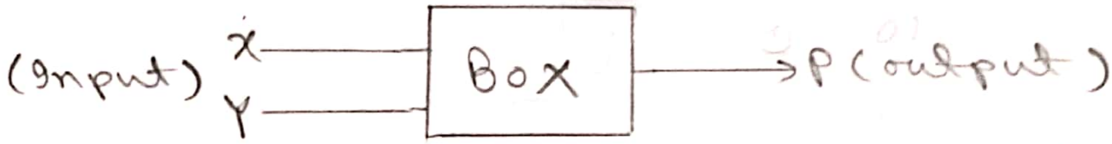
$$= A \oplus (B \oplus C)$$



#

x	
x_1	x_0

Y	
Y_1	Y_0



1. If both strings are in activates.
2. " " " " " deactivates.

→ x_1 x_0 Y_1 Y_0

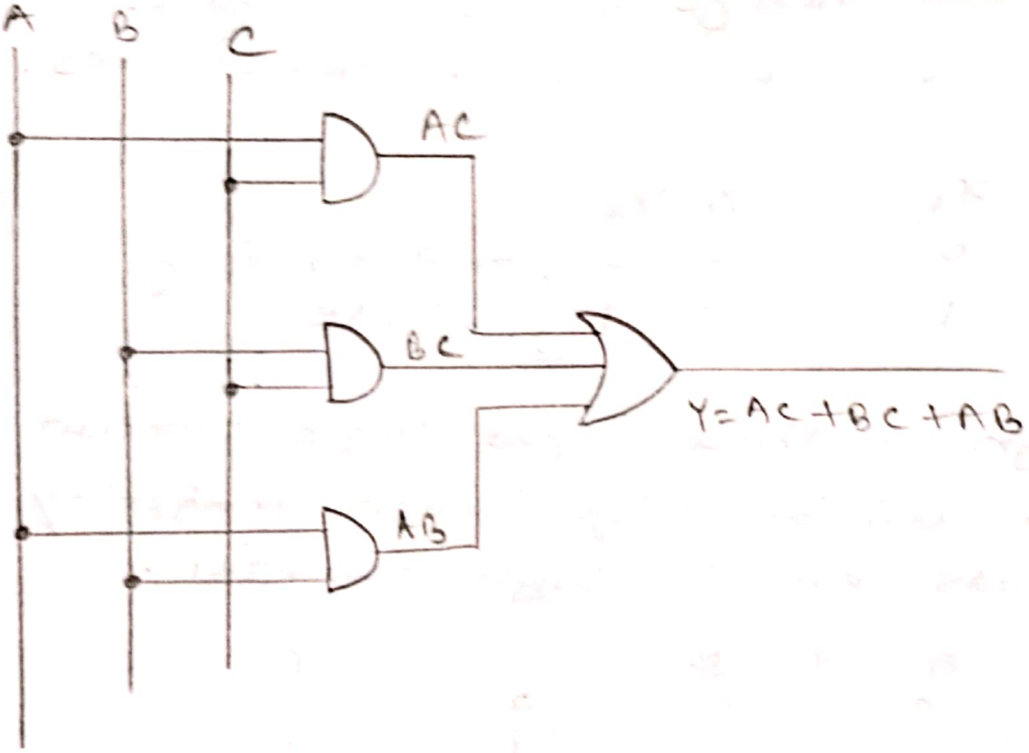
x_1	x_0	Y_1	Y_0	
0	0	0	0	→ Activating state
1	1	1	1	

Design a three input logic circuit whose output will be high when majority numbers of inputs are in high condition.

A	B	C	Y
0	0	0	0
0	1	1	1
1	0	0	0
1	1	1	1
0	0	1	0
0	1	0	0
1	0	1	1
1	1	0	1

BC \ A	0	1
00	0	0
01	0	1
11	1	1
10	0	1

$$Y = AC + BC + AB$$



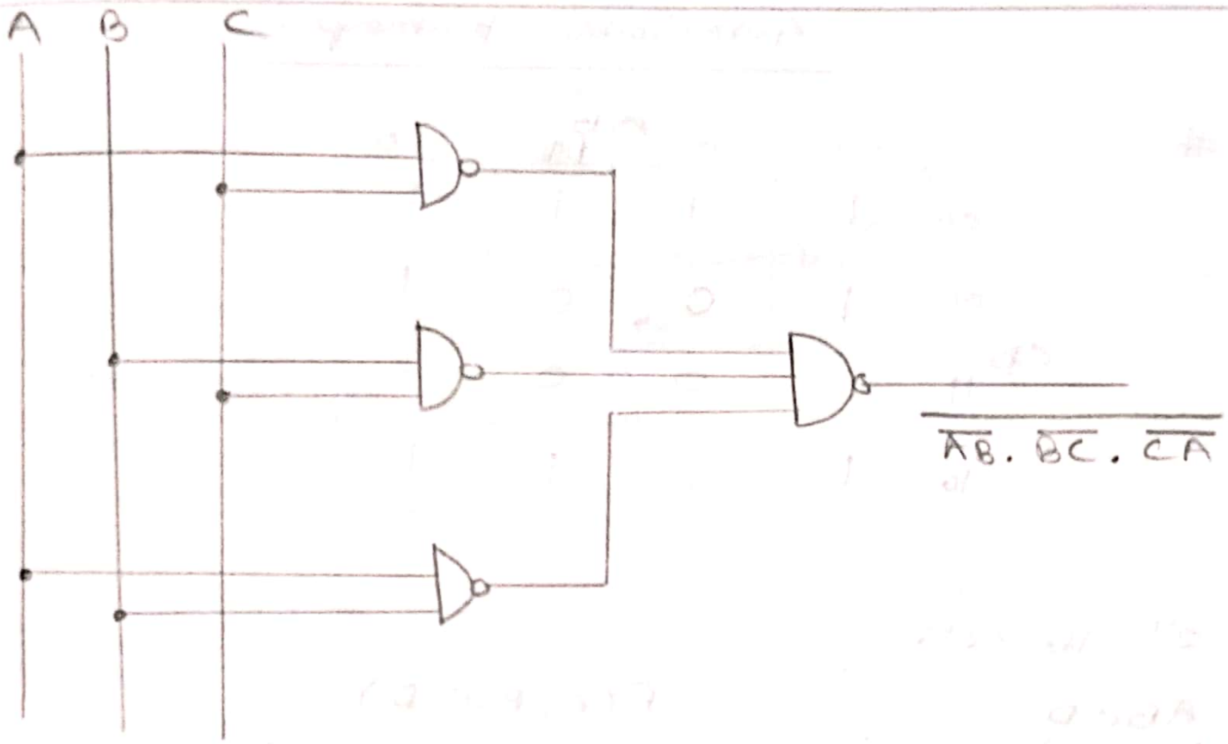
using only universal gate,

$$Y = AC + BC + AB$$

$$= \overline{\overline{AC + BC + AB}}$$

$$= \overline{\overline{AC} \cdot \overline{BC} \cdot \overline{AB}}$$

$\cancel{\overline{AC}}$



using only NOR Gate, (minimum number of gate)

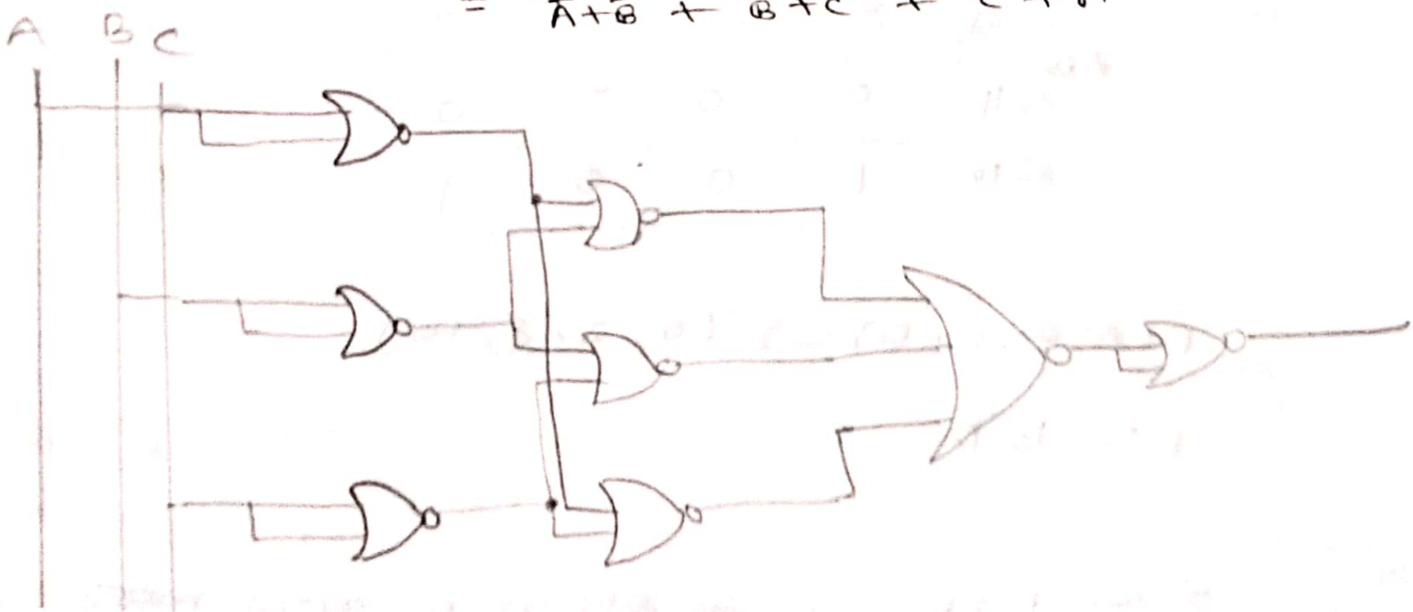
$$Y = AB + BC + CA$$

H.W

$$= \overline{\overline{AB}} + \overline{\overline{BC}} + \overline{\overline{CA}}$$

$$\neq \cancel{\overline{A+B}} \qquad = \overline{\bar{A}+\bar{B}} + \overline{\bar{B}+\bar{C}} + \overline{\bar{C}+\bar{A}}$$

$$= \overline{\overline{\overline{\bar{A}+\bar{B}} + \overline{\bar{B}+\bar{C}} + \overline{\bar{C}+\bar{A}}}}$$



08.11.2022

4-variable K-map

#

CD \ AB	00	01	11	10
00	1_{0}	1_{4}	1_{12}	1_{8}
01	1_{1}	0_{5}	0_{13}	1_{9}
11	1_{2}	0_{7}	0_{15}	1_{11}
10	1_{3}	1_{6}	1_{14}	1_{10}

$2^4 = 16$ cell

ABCD
MSB LSB

$F(A,B,C,D)$
$= \Sigma(0,1,2,3,4,6,8,9,10,11,$
$12,14)$

$F = \overline{B} + \overline{D}$

#

AB \ CD	00	01	11	10
$\bar{A}\bar{B}$ 00	1_{0}	0_{1}	0_{3}	1_{2}
$\bar{A}B$ 01	0_{4}	0_{5}	0_{7}	0_{6}
AB 11	0_{12}	0_{13}	0_{15}	0_{14}
$A\bar{B}$ 10	1_{8}	0_{9}	0_{11}	1_{10}

$F(A,B,C,D) = \Sigma(0,2,8,10)$

$F = \overline{B}\,\overline{D}$

2^n টা 1 যদি loop এর মধ্যে থাকে, তাহলে n টা variable eleminate হয়।

$F(A,B,C,D) = \pi(5,7,13,15)$

↳ maxterm

	C+D			
A+B	0+0	0+1	1+1	1+0
0+0	1 (0)	1 (1)	1 (3)	1 (2)
0+1	1 (4)	0 (5)	0 (7)	1 (6)
1+1	1 (12)	0 (13)	0 (15)	1 (14)
1+0	1 (8)	1 (9)	1 (11)	1 (10)

$F = \overline{B} + \overline{D}$

5-variable k-map (A, B, C, D, E)

A = 0

	DE			
BC	00	01	11	10
00	1 (0)	0 (1)	0 (3)	1 (2)
01	0 (4)	0 (5)	0 (7)	0 (6)
11	0 (12)	0 (13)	0 (15)	0 (14)
10	1 (8)	0 (9)	0 (11)	1 (10)

A = 1

	DE			
BC	00	01	11	10
00	1 (16)	0 (17)	0 (19)	1 (18)
01	0 (20)	0 (21)	0 (23)	0 (22)
11	0 (28)	0 (29)	0 (31)	0 (30)
10	1 (24)	0 (25)	0 (27)	1 (26)

$F = \overline{C}\,\overline{E}$

Don't care function

$F(A,B,C,D)$

$= \Sigma(1,3,5,7,9,13,15)$

$d(A,B,C,D)$

$= \Sigma(0,4,10,11)$

	AB			
CD	00	01	11	10
00	$\cancel{0}$ x (0)	$\cancel{0}$ x (4)	0 (12)	0 (8)
01	1 (1)	1 (5)	1 (13)	1 (9)
10	1 (3)	1 (7)	1 (15)	x 1 (11)
11	0 (2)	0 (8)	0 (14)	0 x (10)

$x \rightarrow 1$ হলে better

don't care এর function এর loop একসাথে করে দেখে না কেন শুধু তার 1 হবে, main func এর কোনো 1 না থাকে।

don't care function's input either be 0 or 1 depending on personal use.

main function কে simply করার জন্য 0/1 ধরতে হবে প্রয়োজন অনুসারে।

প্রথমে d, f দুটোর জন্য একসাথে k-map design করে, নিজে আলাদা k-map draw করতে হবে for looping (don't care এর value বসিয়ে)।

<u>CT-02</u> : lec-4,5 (till don't care)

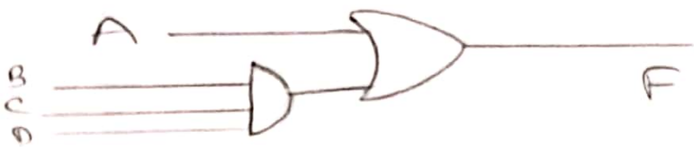
<u>21.11.2022</u>

Design a logic circuit that is to produce a High output when the voltage (represented by a four-bit binary number ABCD) is greater than 6 V.

Voltage				O/P
A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1 → 7
⋮	⋮	⋮	⋮	
1	1	1	1	1

AB \ CD	00	01	11	10
00	0 (0)	0 (1)	0 (3)	0 (2)
01	0 (4)	0 (5)	1 (7)	0 (6)
11	1 (12)	1 (13)	1 (15)	1 (14)
10	1 (8)	1 (9)	1 (11)	1 (10)

$F = A + BCD$



Ex.7 (slide-5)

A	B	C	D	F		
0	0	0	0	1	$\bar{A}\bar{B}\bar{C}\bar{D}$	
0	0	0	1	1	$\bar{A}\bar{B}\bar{C}D$	
0	0	1	0	1	$\bar{A}\bar{B}C\bar{D}$	
0	0	1	1	1	$\bar{A}\bar{B}CD$	
0	1	0	0	1	$\bar{A}B\bar{C}\bar{D}$	
0	1	0	1	0		$A+\bar{B}+C+\bar{D}$
0	1	1	0	0		$A+\bar{B}+\bar{C}+D$
0	1	1	1	0		$A+\bar{B}+\bar{C}+\bar{D}$
1	0	0	0	1	$A\bar{B}\bar{C}\bar{D}$	
1	0	0	1	1	$A\bar{B}\bar{C}D$	
1	0	1	0	0		$\bar{A}+B+\bar{C}+D$

1	0	1	1	0		$\bar{A}+B+\bar{C}+\bar{D}$
1	1	0	0	1	$AB\bar{C}\bar{D}$	
1	1	0	1	0		$\bar{A}+\bar{B}+C+\bar{D}$
1	1	1	0	0		$\bar{A}+\bar{B}+\bar{C}+D$
1	1	1	1	0		$\bar{A}+\bar{B}+\bar{C}+\bar{D}$

$$F=\bar{A}\bar{B}\bar{C}\bar{D}+\bar{A}\bar{B}\bar{C}D+\bar{A}\bar{B}C\bar{D}+\bar{A}\bar{B}CD+$$
$$\bar{A}B\bar{C}\bar{D}+A\bar{B}\bar{C}\bar{D}+A\bar{B}\bar{C}D+AB\bar{C}\bar{D}$$

CD

AB \ CD	00	01	11	10
00	1_{0}	1_{1}	1_{3}	1_{2}
01	1_{4}	0_{5}	0_{7}	0_{6}
11	1_{2}	0_{13}	0_{15}	0_{14}
10	1_{8}	1_{9}	0_{11}	0_{10}

$$F=\bar{A}\bar{B}+\bar{C}\bar{D}+A\bar{B}\bar{C}$$

22.11.23

Simplify the following expression.

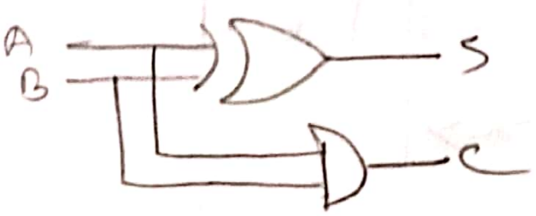
$$f = \overline{AB+AC} + \bar{A}\bar{B}C$$
$$= \overline{AB}.\overline{AC} + \bar{A}\bar{B}C$$
$$= (\bar{A}+\bar{B}).(\bar{A}+\bar{C}) + \bar{A}\bar{B}C$$
$$= \bar{A}\bar{A} + \bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}\bar{C} + \bar{A}\bar{B}C$$
$$= \bar{A} + \bar{A}\bar{B}(1+C) + \bar{C}(\bar{A}+\bar{B})$$
$$= \bar{A} + \bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}\bar{C}$$
$$= \cancel{\bar{A}+}\ \bar{A}(1+\bar{B}) + \bar{A}\bar{C} + \bar{B}\bar{C}$$
$$= \bar{A} + \bar{A}\bar{C} + \bar{B}\bar{C}$$
$$= \bar{A}(1+C) + \bar{B}\bar{C}$$
$$= \bar{A} + \bar{B}\bar{C}$$

Design an adder—

1. Half Adder
2. full Adder

→ 2-bit Half Adder

Input		Output	
A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0



$S = A\bar{B} + \bar{A}B = A \oplus B$

$C = AB$

Half Adder

Full Adder

Input			Output	
x	Y	Z	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$Sum = \Sigma(1,2,4,7)$

$carry = \Sigma(3,5,6,7)$

Sum

$x \backslash YZ$	00	01	11	10
0	0_0	1_1	0_3	1_2
1	1_4	0_5	1_7	0_6

$S = x\bar{y}\bar{z} + \bar{x}\bar{y}z + xyz + \bar{x}y\bar{z}$

Carry

$x \backslash YZ$	00	01	11	10
0	0_0	0_1	1_3	0_2
1	0_4	1_5	1_7	1_6

$C = yz + xz + xy$

CS CamScanner

$$S = x\bar{y}\bar{z} + \bar{x}\bar{y}z + xyz + \bar{x}y\bar{z}$$

$$= x(\bar{y}\bar{z} + yz) + \bar{x}(\bar{y}z + y\bar{z})$$

$$= x(\overline{y \oplus z}) + \bar{x}(y \oplus z)$$

$$= x \oplus (y \oplus z)$$

$\rightarrow$ Full Adder using half Adder / Carry look Ahead Adder

Full Adder,

$Sum = A \oplus (B \oplus C)$

$Carry = AB + BC + CA$

Half Adder,

$Sum = A \oplus B$

$Carry = AB$

$Sum = A \oplus X$ [let, $X = B \oplus C$]

$$Carry = AB + BC + CA$$

$$= AB(C + \bar{C}) + BC(A + \bar{A}) + CA(B + \bar{B})$$

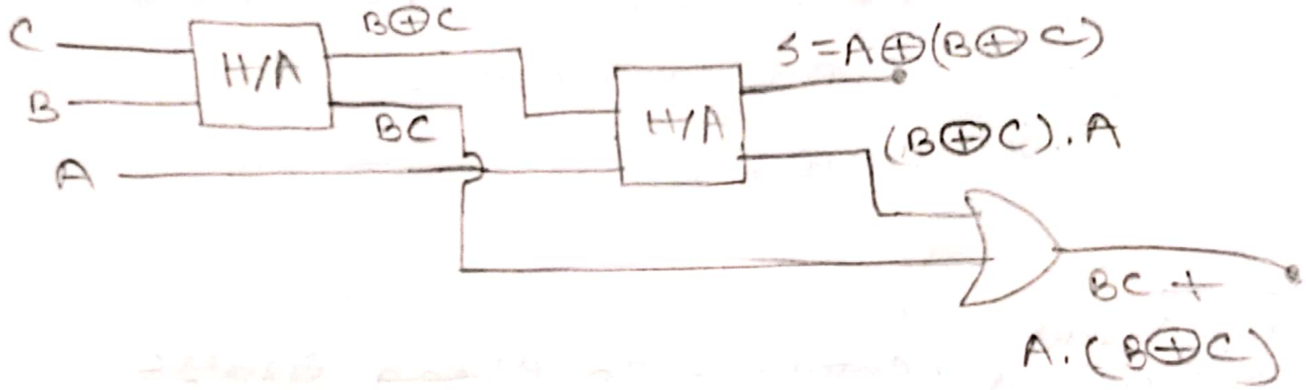
$$= ABC + AB\bar{C} + ABC + \bar{A}BC + ABC + A\bar{B}C$$

$$= ABC + \bar{A}BC + A\bar{B}C + AB\bar{C}$$

$$= ABC + \bar{A}BC + A(\bar{B}C + B\bar{C})$$

$$\cancel{= ABC + \bar{A}BC} = BC(A + \bar{A}) + A(\bar{B}C + B\bar{C})$$

$$= BC + A(B \oplus C)$$

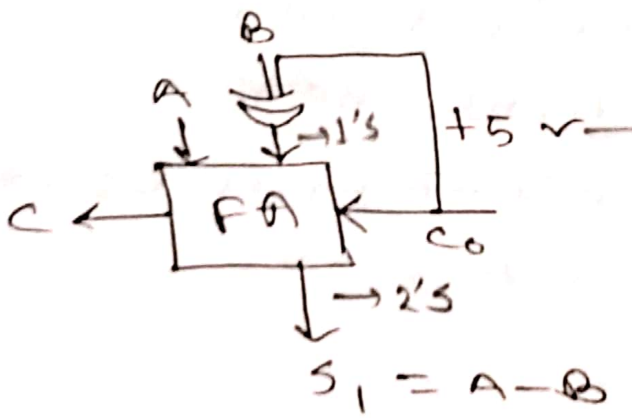


Universal Adder (Addition Subtraction)
(Parallel bit)

$A = A_3 A_2 A_1 A_0$

$B = B_3 B_2 B_1 B_0$

$A - B =$



+5 v → Control input

0 [0 V] ⇒ Add

1 [5 V] ⇒ Sub

28.11.2022

BCD Adder

BCD → four bit representation

523

0101 0010 0011 → $(010100100011)_{BCD}$

23 → 00100011
28 → 00101000

51 → 0100 1011 → greater than 9
$\underbrace{0100}_{4}$ $\underbrace{1011}_{11}$

+0110 → correction factor

$\underbrace{0101}_{5}$ $\underbrace{0001}_{1}$

1001 → 9 → highest decimal
1's → 0110 → 6
1111 → 15 → highest error

238 → 0010 0011 1000
296 → 0010 1001 0110

534

0100 1100 1110
+ 0110 0110

$\underbrace{0101}_{5}$ $\underbrace{0011}_{3}$ $\underbrace{0100}_{4}$

6 → 0110
4 → 0100

10

1010 → >9
+0110

1 0000

Book: Tocci, Morison